

# VerusCite V1 Benchmark: Citation Verification Accuracy

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## 1 The Problem

Fabricated citations in academic papers have grown sharply since the widespread adoption of generative AI writing tools. A Lancet audit of 2.5 million biomedical papers found hallucinated

reference rates rising 12-fold between 2023 and early 2026 (Topaz et al., 2026). GPTZero’s audit of NeurIPS 2025 found over 100 confirmed hallucinated citations across 53 accepted papers despite rigorous peer review. Separate estimates project roughly 147,000 hallucinated citations across arXiv, bioRxiv, SSRN, and PMC in 2025 alone.

Existing peer review and editorial safeguards largely fail to catch these errors before publication. The problem spans disciplines: biomedical research, AI/ML conferences, legal filings, and government reports.

VerusCite ([veruscite.com](https://veruscite.com)) is a tool to automatically extract and verify citations from uploaded documents. This report documents the accuracy of the V1 verification pipeline against a hand-labeled ground truth corpus.

## 2 Approach

### 2.1 Design Philosophy

Even in documents with hallucinated citations, the majority of references are legitimate. A tool with a high false positive rate – flagging real papers as hallucinations – would waste a reviewer’s time chasing false alarms rather than finding actual problems. VerusCite prioritizes **precision over recall**: it is better to miss some hallucinations than to incorrectly flag verified citations.

The target false positive rate for hallucination flags is below 1%. Recall for detecting actual hallucinations is around 70-80%. This tradeoff means a human reviewer spends most of their time on citations that genuinely need attention rather than dismissing false reports.

The application is designed for human-in-the-loop review. Each citation’s verification status can be manually overridden, and the interface surfaces the reasoning behind each determination so reviewers can make informed decisions quickly.

### 2.2 Comparison to Similar Tools

This approach is similar to [Pangram](#), which also emphasizes low false positive rates for AI-generated content detection. Their published materials discuss the same tradeoff: when base rates of problematic content are low, even small false positive rates cause most flagged items to be false alarms (see [Pangram’s blog](#)).

### 3 Pipeline Architecture

VerusCite processes documents in two stages:

1. **Citation Extraction** – An LLM reads the document text and identifies individual references, parsing out structured fields (title, authors, year, journal, DOI).
2. **Citation Verification** – Each extracted citation is checked against external sources to determine whether it exists as described.

#### 3.1 Verification Categories

The checker assigns each citation one of four statuses:

| Status               | Meaning                                                                           |
|----------------------|-----------------------------------------------------------------------------------|
| <b>Verified</b>      | Citation confirmed to exist via CrossRef or web search                            |
| <b>Minor Error</b>   | Citation likely exists but has metadata discrepancies (wrong year, volume, pages) |
| <b>Hallucination</b> | No evidence the cited work exists as described                                    |
| <b>Not Found</b>     | Unable to confirm or deny (insufficient search results)                           |

#### 3.2 Verification Method

Verification proceeds in two passes:

1. **Static CrossRef lookup** – A title/DOI search against CrossRef metadata. Fast and free. Confirms roughly 60% of legitimate citations without needing web search.
2. **Web search verification** – Citations not resolved by CrossRef are sent to a web-search-enabled LLM (Perplexity or OpenAI web search tools) to locate evidence of the publication.

The primary models used are **Perplexity** (hosting Gemini Flash Lite or Gemini 3 Flash) and **OpenAI GPT-5.4-mini/nano** as fallback. Having multiple providers is important operationally – any single provider experiences periodic outages or rate limits. Results reported here represent the range across these configurations.

Cost is kept low by using smaller models with web search rather than large frontier models. A full corpus check (2,200+ citations) runs between \$15-\$44 depending on the model.

## 4 Ground Truth

The V1 validation corpus contains **2287** hand-labeled citations across **36 documents**. Sources are intentionally heterogeneous:

- Papers with known hallucinations identified by others on social media (Chris Carothers, David Buil-Gil)
- GPTZero’s NeurIPS 2025 audit ([gptzero.me/news/neurips](https://gptzero.me/news/neurips))
- Papers flagged by [Reviewer3](#) (see also [@Reviewer3 on X](#))
- A ChatGPT deep-research output where all 12 citations are fabricated
- Clean papers (dissertation excerpts, open-access criminology, PLoS ONE articles)
- MDPI and preprint samples

Label distribution:

- Verified: 1945 (85.0%)
- Minor Error: 148 (6.5%)
- Hallucination: 140 (6.1%)
- Not Found: 54 (2.4%)

The corpus skews heavily toward verified citations (as real documents do), which makes false positive rate the critical metric.

## 5 Extraction Results

Citation extraction uses an LLM to parse the document text and identify individual references. The table below shows extraction accuracy against the ground truth.

```
ext = extraction_table()
ext
```

Table 2: Extraction accuracy by model and OCR backend

|   | Model                 | OCR       | Real | Correct | Missing | Extra | Rate (%) | Cost (\$) |
|---|-----------------------|-----------|------|---------|---------|-------|----------|-----------|
| 0 | gemini-3.1-flash-lite | pypdfium2 | 2287 | 2284    | 3       | 12    | 99.9     | 1.156257  |
| 1 | gpt-5.4-nano          | pypdfium2 | 2287 | 2282    | 5       | 13    | 99.8     | 0.882243  |

Extraction is very accurate across models. Gemini 3.1 Flash Lite with pypdfium2 correctly extracts over 99.8% of citations at under \$1.20 for the full corpus. The few “extra” rows are

typically page footers, footnotes, or duplicate listings that the LLM picks up in addition to the actual bibliography.

The `liteparse` OCR backend (a lighter PDF text extractor) produces slightly more misses than `pypdfium2`, which handles multi-column layouts and ligatures better.

## 6 Checker Results

```
chk = checking_table()
chk
```

Table 3: Checker accuracy by model/provider

|   | Model                         | Provider   | FP → Hallucination | FP → Minor Error | Hallucination Recall |
|---|-------------------------------|------------|--------------------|------------------|----------------------|
| 0 | google/gemini-3-flash-preview | perplexity | 0.1% (1)           | 1.8 (36)         | 73.6% (103/140)      |
| 1 | google/gemini-3.1-flash-lite  | perplexity | 0.4% (8)           | 2.5% (49)        | 70.7% (99/140)       |
| 2 | gpt-5.4-mini                  | openai     | 0.1% (2)           | 3.6% (71)        | 61.4% (86/140)       |
| 3 | gpt-5.4-nano                  | openai     | 0.3% (6)           | 4.3% (83)        | 57.9% (81/140)       |

Key metrics:

- **FP → Hallucination:** Percentage of actually-verified citations incorrectly flagged as hallucinations. All configurations stay at or below 0.4%.
- **Not-Verified Recall:** Percentage of actual problems (hallucinations + minor errors + not-found) correctly identified. Ranges from 72-82%.
- **Hallucination Recall:** Percentage of actual hallucinations correctly flagged. The best configuration (Gemini 3 Flash via Perplexity) achieves 73.6%.

### 6.1 Cost Breakdown

```
costs = cost_comparison_table()
costs
```

Table 4: Cost breakdown per full-corpus checker run

|   | Model                         | Provider   | Token Cost (\$) | Search Cost (\$) | Total (\$) |
|---|-------------------------------|------------|-----------------|------------------|------------|
| 0 | google/gemini-3-flash-preview | perplexity | 17.129258       | 4.680            | 21.809336  |
| 1 | google/gemini-3.1-flash-lite  | perplexity | 9.578944        | 5.195            | 14.773963  |
| 2 | gpt-5.4-mini                  | openai     | 25.198293       | 18.100           | 43.298306  |
| 3 | gpt-5.4-nano                  | openai     | 7.046987        | 16.480           | 23.526988  |

Perplexity-hosted models are substantially cheaper than direct OpenAI for equivalent or better accuracy. The Gemini 3.1 Flash Lite configuration provides the best cost/accuracy ratio at ~\$15 per full run.

## 6.2 CrossRef as First Pass

Across all checker runs, approximately **1,331** of ~1,946 verified citations are confirmed via static CrossRef lookup alone (no web search needed). This is roughly 68% of verified citations resolved without any LLM cost, which keeps per-citation expenses low and latency down.

## 7 Limitations and Next Steps

The V1 corpus is heterogeneous by design, but 36 documents is a limited sample. The next evaluation (V2) will use the same prompts and pipeline but a **completely different validation set** – different documents, different domains, different sources of known hallucinations. This ensures the system is not overtrained on the particular examples in V1.

Known limitations of the current evaluation:

- Ground truth labels for some documents were seeded from the checker itself and hand-reviewed, which may introduce subtle bias toward the checker’s own patterns.
- “Not found” is an inherently ambiguous category – some citations exist but are difficult to locate via web search (paywalled, very recent, or in non-English databases).
- The corpus over-represents arXiv preprints relative to other domains (law, humanities, clinical medicine).

## 8 Reproducibility

All data for this report is public at [github.com/apwhee/veruscite-data](https://github.com/apwhee/veruscite-data). The v1/ directory contains:

- `ground_truth.csv` – hand-labeled citations with `expected_status`
- `extraction_run/` – raw extraction outputs per model
- `checking_run/` – raw checker outputs per model
- `manifest.json` – run IDs used in this report

To regenerate this report:

```
pip install -r requirements.txt  
quarto render report.qmd
```